Appendix 3: Canada's Energy Policies – Sustainable Development

Each of us uses energy everyday. We use energy to light and heat our homes; to cook our meals. We use energy to travel to and from our workplaces and we use energy, indirectly, in the goods we purchase and the services provided to us. Energy is an integral part of our everyday lives. How we produce and consume that energy affects our economy, demonstrates our responsibility to the environment and future generations and helps to define us as a society.

Canadians are fortunate to have secure, reliable and diverse sources of energy. We have ample supplies of fossil fuels, hydro-electric power, uranium and coal. Although still a relatively small source compared to traditional fuels, alternative forms of energy, such as ethanol, wind-power, and solar energy are forming a growing part of the energy mix available to Canadians.
The production and consumption of energy affects our environment. The burning of fossil fuels releases carbon dioxide and other air pollutants; the damming of large rivers for hydro-electric power results in flooding; and, waste from nuclear energy production has long-term disposal issues to be resolved. In 1987, The World Commission on Environment and Development (The Bruntland Commission) introduced the phrase "sustainable development" which it defined as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs".

Today, Canadian energy policy is oriented toward this concept of sustainable development. Canadian energy policy is not simply concerned with production and supply issues. The federal government, primarily Natural Resources Canada, employs a range of policies and programs to address specific energy issues within the overall context of a market-based system of energy production and pricing.

Conventional energy sources, defined as fossil fuels and electricity generated from oil, gas, coal and nuclear power plants, remain the dominant energy forms in Canada. Conventional energy sources contribute over 7% to our Gross Domestic Product, account for annual investment of approximately $24 billion and result in the direct employment of nearly 280,000 people. Fossil fuels are produced, primarily, from Canada's Western Sedimentary Basin and offshore on its east-coast. The nuclear power industry, fueled by domestic uranium, operates 22 Candu reactors in Canada and exports its Candu technology around the world. Coal, mined domestically, is the primary fuel for electricity generation in a number of provinces.

Part of the sustainable development concept is the efficient use of energy - not wasting valuable resources. Another element is the adoption of alternative sources of energy Canada is a leader in energy efficiency technologies and research and development of renewable and alternative energy sources, including hydraulic, solar, wind and biomass and innovative technologies, such as the fuel cell. In order to foster these technologies, Natural Resources Canada delivers efficiency and renewable energy programs to virtually all sectors of the economy to educate and demonstrate to end-users the opportunities available to them to reduce energy consumption and protect the environment. The traditional instruments of government, regulation and taxation, are also employed, as is direct support for research and development the programs such as Program on Energy Research and Development (PERD). As a result, Canada is a leader in energy research and innovative technology development and exports that expertise around the world.

Canada, as an actor on the international stage, has responsibilities to the global community and works with other nations to address issues of trade and global concern. Energy, being an internationally traded commodity and emissions from its use not respecting national boundaries, is the subject of many international agreements. These agreements, whether addressing trade standards or transboundary air pollution, have a great influence on how we develop and carry out energy policy in Canada. One such agreement, The Kyoto Protocol to the United Nations Framework Convention on Climate Change, signed in December 1997 by 160 countries including

Canada, will have a significant impact on Canadian energy policy. Under Kyoto, Canada agreed to a reduction in its greenhouse gas (GHG) emissions of 6% from 1990 levels by the period 2008 to 2012. As nearly 90% of all anthropogenic GHG emissions in Canada result from the production and consumption of fossil fuels, achieving this goal will challenge all Canadian governments and the energy industry to develop effective policies and programs to limit GHG emissions.

Energy policy in Canada must, therefore, reflect a balance of these issues - an economically competitive and innovative energy sector that contributes to the wealth of our society; a secure, reliable and safe supply of energy for all; energy production and use that respects the environment and that is sustainable for future generations; and, Canada's responsibility acting within the community of nations to resolve global issues.

http://www.nrcan.gc.ca/es/epb/eng/sustainable_e.htm